From Kaggle to Causal Inference

A short project guide for M2 Causal Inference students

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Kaggle datasets are built for prediction (*what will Y be?*). Your project turns one into a causal question (*how much does Y change if we change T?*). The hard part is not the download: it is finding a credible treatment, a clear outcome and an argument for why the comparison is fair.

1 Getting the data

Download the CSV directly from the dataset page on kaggle.com/datasets (*Download* button), put it next to your notebook, and load it with pandas. Record the dataset URL and the download date in the first cell of your notebook.

```
import pandas as pd
df = pd.read_csv("StudentsPerformance.csv") # file downloaded from Kaggle
```

A dataset is usable only if it has a **treatment** someone could switch on or off (a course, an ad, a price, a policy — never age, gender or ethnicity), an **outcome measured after it**, pre-treatment **confounders**, and a documented **assignment story**: how did units end up treated?

2 The workflow in seven steps

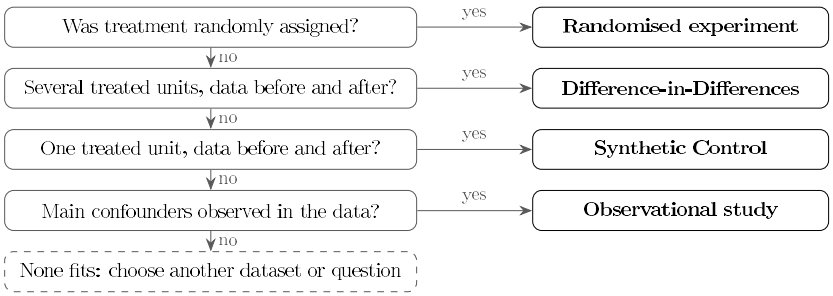
Step	What to do	Output in the notebook
1. Question	Name the unit, the treatment T, the outcome Y and the effect you want (ATE, ATT or ATU)	One sentence: "What is the effect of [T] on [Y]?"
2. Variables	Say which variables are confounders (they affect both T and Y) and which are measured after T (never control for those)	A short table of variables and their role
3. Method	Pick the method from how the treatment was assigned (diagram below)	Method + its key assumption
4. Compare the groups	Compare treated and control units before treatment: are they similar?	Descriptive table and plots
5. Estimate	Start with the naive difference, then the estimate from your method	Effect with a confidence interval
6. Check	Test the result: does it change when you change a choice (another control variable, a subset of the data, a fake treatment date)?	Short list of checks and what they show
7. Conclude	Answer the question, explain the limits, give a recommendation	Conclusion

$$\text{ATE} = \mathbb{E}\big[Y_i(1) - Y_i(0)\big] \qquad \text{ATT} = \mathbb{E}\big[Y_i(1) - Y_i(0) \mid T_i = 1\big] \qquad \text{ATU} = \mathbb{E}\big[Y_i(1) - Y_i(0) \mid T_i = 0\big]$$

Example (warm-up). In *Students Performance in Exams*: T = test preparation course, Y = math score, confounders = parental education and lunch (income proxy). Do **not** control for reading and writing scores: they are also outcomes of the course.

Choosing the method: ask in order, the first "yes" wins

Use only the methods seen in class: randomised experiments, observational studies, Difference-in-Differences and Synthetic Control.



Method	How the effect is computed	Key assumption
Randomised experiment	Mean of treated − mean of control	Treatment was really random
Observational study	Compare treated and control units that have the same confounders (e.g. regression with the confounders)	All important confounders are observed
Difference-in-Differences	(change over time for treated) − (change over time for control)	Parallel trends without treatment
Synthetic Control	Treated unit compared with a weighted mix of untreated units that matched it before treatment	The mix tracks the treated unit well before treatment

Simple code examples

```
import statsmodels.formula.api as smf

# Randomised experiment / naive difference
naive = df[df.T == 1].Y.mean() - df[df.T == 0].Y.mean()

# Observational study: compare at equal confounders
obs = smf.ols("Y ~ T + parent_edu + lunch", data=df).fit()
print(obs.params["T"], obs.conf_int().loc["T"])

# Difference-in-Differences (2 groups x 2 periods)
m = df.groupby(["treated", "post"]).Y.mean()
did = (m[1, 1] - m[1, 0]) - (m[0, 1] - m[0, 0])
```

3 Your project

- **Groups** of 2–3 students, one Kaggle dataset per group (no two groups on the same dataset).
- **Deliverables:** (1) a **written report of about 10 pages** (PDF) presenting all of the work, (2) the **Jupyter notebook** (runs top to bottom on a fresh Colab), and (3) a **10-minute presentation.**
- **Workplace mirror:** one paragraph linking your question to a decision at your host company. Kaggle data only — no company data.
- **Avoid** prediction classics (Titanic, Iris, house prices). Simulated datasets are allowed if you say so.

Deadline: the project (10-page report + notebook) must be submitted **one week before the oral presentation.** No changes are accepted after that date.

Structure of the 10-page report

The report must be understandable without opening the notebook. Figures and tables come from the notebook; code does not go in the report.

Section	Pages	Content
1. Introduction	1	Context, the causal question, treatment, outcome, ATE / ATT / ATU, link to your company
2. Data	1	Kaggle source, variables, how the treatment was assigned
3. Causal framework	1.5	Potential outcomes, the effect you estimate, the causal graph, confounders
4. Method	1.5	The method chosen (randomised experiment, observational study, DiD or Synthetic Control) and why; its key assumption
5. Descriptive analysis	1.5	Naive difference; comparison of treated and control groups (or trends over time)
6. Results	2	The estimated effect with a confidence interval and its interpretation
7. Checks and limits	1	What could make the result wrong; checks you ran
8. Conclusion	0.5	Answer and recommendation

Evaluation

Component	Weight	What is evaluated
Quality of the analysis	**45%**	Clear causal question and estimand (10%), correct choice of variables and confounders (10%), method adapted to the data and assumption discussed (15%), correct estimation and interpretation (10%)
Quality of the report	**20%**	Structure and clarity (10%), discussion of limits (5%), recommendation and link to the company (5%)
Quality of the presentation	**20%**	Clear storyline within 10 minutes (8%), slides and visuals (6%), answers to questions (6%)
Quality of the code	**10%**	Notebook runs end to end (5%); readable and commented (5%)
Respect of the deadline	**5%**	Project submitted one week before the oral presentation
Total	**100%**	

Dataset ideas

Kaggle handle	Treatment → outcome	Design
`mursideyarkin/mobile-games-ab-testing-cookie-cats`	Gate 40 vs 30 → retention	Randomised
`chebotinaa/fast-food-marketing-campaign-ab-test`	Promotion → sales	Randomised
`spscientist/students-performance-in-exams`	Prep course → math score	Observational
`pavansubhasht/ibm-hr-analytics-attrition-dataset`	Overtime → attrition	Observational
`blastchar/telco-customer-churn`	Tech support → churn	Observational

Most common mistakes: controlling for variables measured after the treatment; reading a correlation as a causal effect; not saying for whom the effect holds (ATE, ATT or ATU); DiD without checking trends before treatment.